

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE
2026

EconophysicsLab offsite 2026 | Slide 1

The statistics of soaring flight

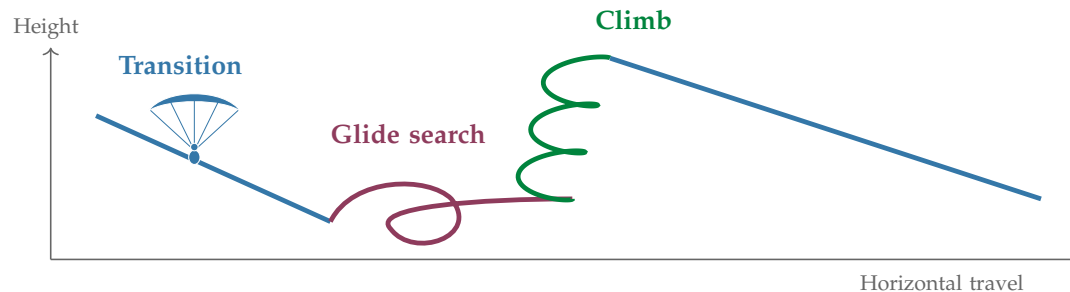
The aim is to describe soaring flight as a transport process and to identify empirical constraints for a later stochastic model. Today I cover the archive and its preparation, horizontal mean-square displacement and conditional comparisons through Section 3.2, followed by open questions and next steps.

The cover evokes the offsite setting: Lake Maggiore, Isola dei Pescatori and the Alps. A glide leads into local search and a helical climb in a thermal rising from a sun-warmed mainland slope. Upward filaments indicate airflow; the white helix is the aircraft path, with wind drift. The foreground island includes its bell tower and a red waterfront building without signs. This AI-generated illustration is geographically inspired, not a literal topographic map, measured airflow or a recorded flight.

Visual references: the user-supplied photograph and Comune di Verbania, Isola Superiore or Isola dei Pescatori, including the aerial photograph credited to Robin Ulrich.

Soaring is transport with an energy budget

Height is spent to travel and recovered in rising air.

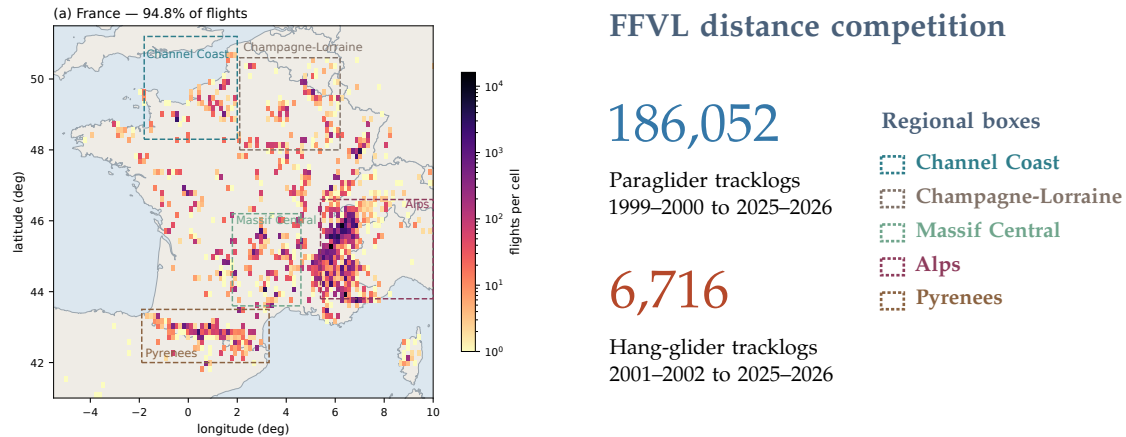


Wind, terrain and route choice act throughout the flight.

A flight is sustained by an energy budget: the pilot trades altitude for horizontal distance, searches for rising air, and recovers altitude in lift. These are useful behavioural descriptions, not phase labels supplied by the recorder or results of a segmentation shown here. Ground motion includes both air-relative motion and wind. The challenge is to explain whole-flight transport while respecting these physical constraints. The present measurements are the first layer of that programme. An MSD exponent alone cannot distinguish all the candidate mechanisms: persistent motion, changes of behaviour and heterogeneous conditions can produce similar second moments. The schematic is illustrative and its horizontal axis has no numerical scale. Source: thesis introduction, Sections 1.1–1.3.

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

France map: 148,081 paraglider and 5,894 hang-glider flights. Colour: flights per 0.15° cell.

The source is the FFVL Coupe Fédérale de Distance, with separate archives for paragliders and hang gliders. The verified downloads contain 186,052 and 6,716 tracklogs. Season names refer to their starting year. The IGC files supply time-stamped coordinates and altitude channels, while the catalogue supplies flight date, declared take-off site, route class and wing class. The map shows the retained population, not the raw downloads; both disciplines are pooled. About 94.8 percent of retained flights are in metropolitan France, with a strong Alpine concentration. The coloured boxes are the longitude–latitude windows used for regional comparisons, listed on the right in the same colours: Channel Coast, Champagne-Lorraine, Massif Central, Alps and Pyrenees. Additional flights occur in La Réunion and elsewhere. This is a submitted cross-country archive with uneven geographic coverage, not a random sample of all free flight. Sources: thesis Chapter 2; generated/stats.tex, pipeline_census.tex and prelim_map.pdf.

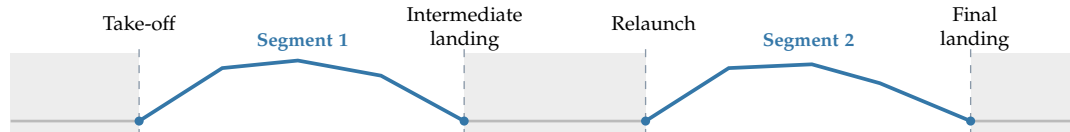
1 GNSS altitude

GNSS height $\longrightarrow$ Trajectory $z(t)$

Barometer $\longrightarrow$ Cleaning checks

GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

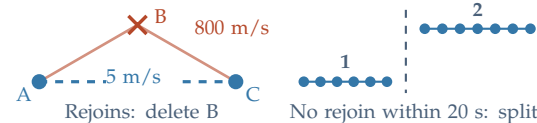
3 Airborne interval



Trim before take-off and after landing.
Stops are not recorded: we infer them from position and altitude.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



● Retained flight

■ Ground removed

Preprocessing: clean the recorded flight

GNSS height is the chosen output channel, used throughout. Its absolute reference needs no pressure or temperature calibration, whereas pressure altitude shifts with the weather. In supported paired records, high-frequency median spectra are comparable to barometry. This does not make GNSS uniformly more accurate: logged vertical datums (ellipsoid or geoid) remain unaudited. Barometry corroborates cleaning decisions without replacing the output altitude.

A fix records time, position and altitude at one instant. The rule is conservative: a fix is deleted only when the flight without it is physically consistent. In the left example, A is at 0 m at 0 s, B at 800 m at 1 s, and C at 10 m at 2 s. Reaching B from A requires an impossible 800 m/s. Removing B leaves 10 m in 2 s: 5 m/s. Deletion therefore passes both the impossibility and rejoin tests. The dashed chord tests reachability; it is not interpolation. Contiguous blocks are handled the same way. In the right example the track jumps and stays displaced: no later fix is reachable from the last accepted one within 20 seconds, the Hampel half-window. Unlike the single spike on the left, the displacement persists past the window, so it cannot be an isolated error. The pipeline then keeps both branches, segments 1 and 2, and splits the flight at the jump without deleting any fix, because it cannot tell which branch holds the true positions. A Hampel flag alone never deletes a fix; timestamp, frozen-lock and altitude defects have separate rules.

The timeline distinguishes take-off, intermediate landing, relaunch and final landing. Grey ground

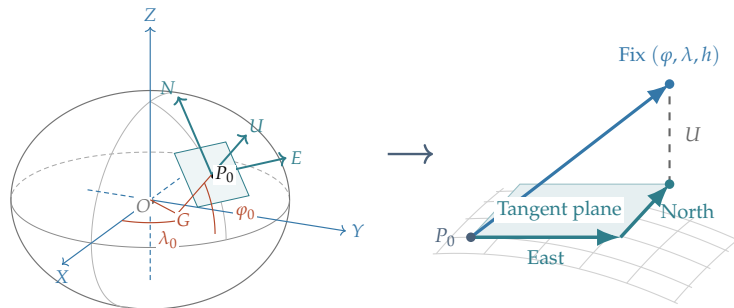
4 Flight selection

Minimum flight thresholds

Time ≥ 40 min1.13% excluded ($N = 191,108$)Path ≥ 20 km0.09% excluded ($N = 188,942$)Altitude range ≥ 600 m2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred position
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid, not a sphere (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

Preprocessing: selection and local coordinates

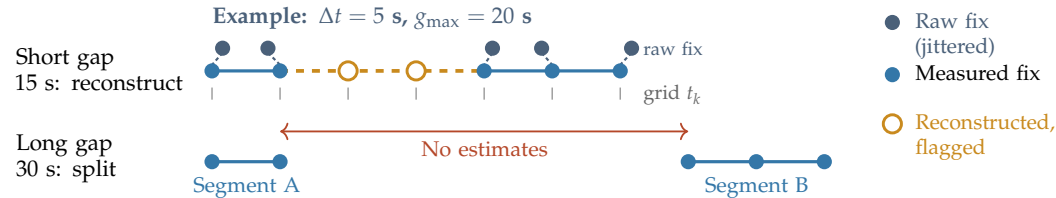
The minima are 40 minutes, 20 kilometres and 600 metres of altitude range. Elapsed span is capped at 16 hours, with additional integrity and reach checks. Each bar pools both disciplines and represents the logs reaching that sequential check. Duration rejects 2,163 of 191,108 (1.13 percent), path rejects 174 of 188,942 (0.09 percent), and altitude range rejects 5,054 of 188,768 (2.68 percent). Three overlong logs leave before the path check. Blue marks the share passing that check. These are cascade rates, rather than independent tests on the raw archive.

Latitude, longitude and logged GNSS height give an Earth-centred Earth-fixed position. The conversion uses the WGS84 ellipsoid rather than a spherical Earth, because GPS positions are expressed in that reference: semi-major axis 6,378,137 m, flattening $1/298.257223563$. Subtract the first trimmed fix, P_0 , then rotate into local east, north and up. The global schematic preserves the ellipsoid normal with exaggerated flattening. The close-up projects a later fix onto the fixed tangent plane. Keep the same origin and orientation for every segment of that flight. P_0 is near, but need not coincide with, true take-off.

The output vertical coordinate remains logged GNSS height, without subtracting origin height. Rotated Up contains curvature and is not used for vertical dynamics. Final retained counts come after stage 7, on the next slide. Sources: thesis Sections 2.7–2.8, local ENU figure and pipeline_census.tex.

6 Time grid

- Put each flight on an exact grid $t_k = t_0 + k\Delta t$ at its own native interval Δt .
- A fix within $\Delta t/2$ of t_k : use it.
 - None that close: reconstruct t_k .
- Reconstruct only a gap $g \leq g_{\max} = \min(10\Delta t, 20 \text{ s})$.
A longer gap splits the flight into segments.



Keep a segment only if it lasts $\geq 90 \text{ s}$ and $\leq 10\%$ of its grid is reconstructed.
No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

Preprocessing: uniform time grid

Stage 6 puts each flight on an exact uniform grid, t_0 plus k times its native interval (the median recording interval). A grid time with a fix within half a step counts as measured; one without is reconstructed (PCHIP horizontally, linear in altitude) and flagged. Jitter is thus removed while the flight keeps its own cadence. At a 5-second cadence the gap cap is 20 seconds. A 15-second gap has two internal grid fixes, linearly reconstructed and flagged: orange dashes and hollow markers. Blue markers are retained fixes in this aligned example. A 30-second gap leaves separate segments with no estimates between them. Mandatory cleaning boundaries always split. The same cap applies between finite altitudes: long altitude holes split the full trajectory. Reconstructed grid fixes are 0.58 percent for paragliders and 0.30 percent for hang gliders. Each resulting segment must last at least 90 seconds and have at most 10 percent reconstructed grid times; a failing segment is dropped alone. A flight with no surviving segment is dropped: 20,798 paraglider and 193 hang-glider flights, the largest single loss in the pipeline. Retained counts follow on the next slide, after stage 7. Sources: thesis uniform-sampling section; resample.py.

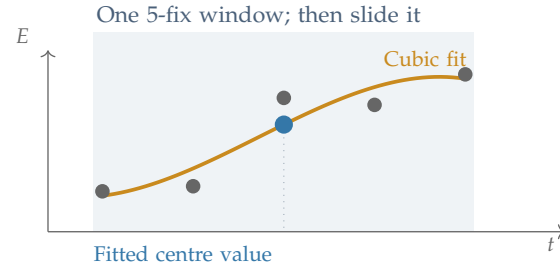
7 Savitzky–Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

Window: odd, ≥ 5 fixes.

Differentiate the polynomial analytically: no finite differences.

$$\frac{d^n \hat{x}}{dt^n}(t_i) = \left. \frac{d^n P_i}{du^n} \right|_{u=0}, \quad u = t - t_i$$



Retained flights after all seven stages (share of downloaded logs)

156,406 paragliders 84.1%

6,094 hang gliders 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

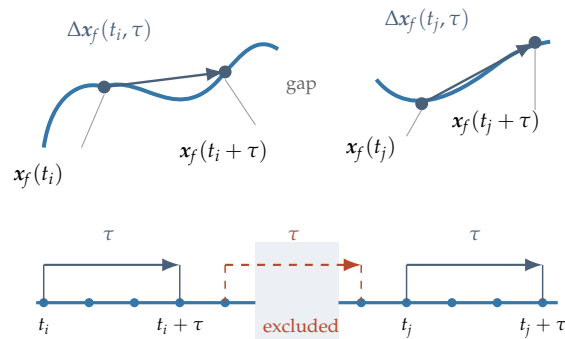
Preprocessing: smoothing and derivatives

Stage 7 fits a cubic independently to E , N and z in a moving window. With physical time offset $u = t - t_i$, the fitted value and successive polynomial derivatives at zero give position, velocity and acceleration. Derivatives through order three are available from a cubic; the pipeline exports the first two. No subsequent finite differences are taken: those would amplify high-frequency position noise. The grey synthetic fixes and orange curve show an exact least-squares fit; the blue dot marks its centre value; the derivatives are those of the orange cubic at that time.

The 5-second configured timescale is converted to an odd window with at least five fixes. Five fixes span 4 seconds at 1-second cadence but 20 seconds at 5-second cadence. Smoothing can therefore affect short-lag statistics differently across cadences. Windows never cross a split; one-sided edge estimates and reconstructed support remain flagged. Retained counts cover the full pipeline: 156,406 of 186,052 paraglider logs and 6,094 of 6,716 hang-glider logs. Stage 6 causes most of the loss. Sources: thesis smoothing section; smoothing.py.

Measuring horizontal transport

One flight f , two retained segments



- The solid arrows (Δx_f) lie in different segments, but each lies entirely within one: $i, j \in \mathcal{I}_f(\tau)$.
- The dashed arrow crosses the gap and is excluded.

For each lag τ :

- 1 Slide the origin t_i through each segment of f , one grid step at a time, while $t_i + \tau$ stays inside
 $\Delta x_f(t_i, \tau) = x_f(t_i + \tau) - x_f(t_i)$
- 2 Pool these origins over all segments of f
 $\mathcal{I}_f(\tau) = \{i : t_i, t_i + \tau \text{ in the same segment}\}$

- 3 Average the $n_f(\tau) = |\mathcal{I}_f(\tau)|$ squared displacements

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta x_f(t_i, \tau)|^2$$

- 4 Average over the $N(\tau)$ flights with at least one origin, each with weight $1/N(\tau)$

$$M_2(\tau) = \frac{1}{N(\tau)} \sum_{f: |\mathcal{I}_f(\tau)| > 0} m_f(\tau)$$

The right column is the computation at one fixed lag τ ; we repeat it for every lag. Step 1: in each segment of flight f , the first origin is the segment start; we move the origin one grid step at a time up to the last point for which $t_i + \tau$ is still in the same segment, so displacements overlap. In the drawing, i and j are two origins of the same flight with the same lag, in different segments. Step 2: pooling over all segments gives $\mathcal{I}_f(\tau)$. No displacement crosses a gap: the dashed arrow has both endpoints in retained segments but is excluded.

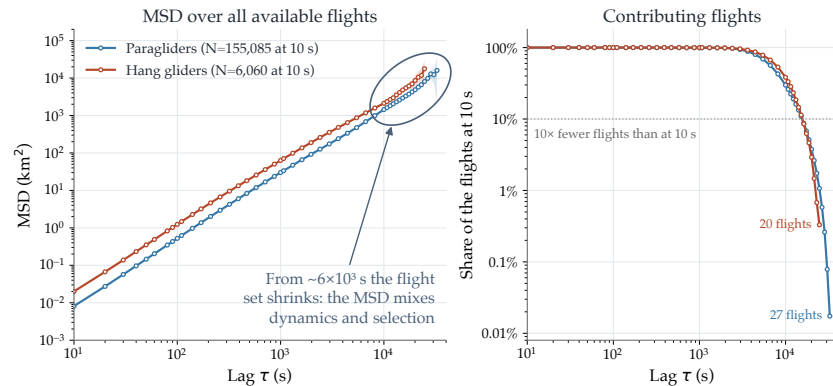
Step 3: the flight has $n_f(\tau) = |\mathcal{I}_f(\tau)|$ displacements, the number of its origins that admit lag τ . Their plain mean square is the time-averaged MSD $m_f(\tau)$. Step 4: the plain mean of $m_f(\tau)$ over the $N(\tau)$ flights with at least one admissible origin; the thesis calls this set $\mathcal{C}(\tau)$. A flight with many origins or segments weighs the same as a flight with few: this is the equal-flight average. Displacement is a chord, not distance travelled.

In the main analysis the grid step is ten seconds, and the flights and segments are the fixed C10000 cohort introduced two slides ahead: only segments of at least 12,500 seconds enter, at every lag. Slide 11 extends the short-lag view with cadence-limited admission. H is half the log-log slope: an effective growth exponent, not an identified self-similarity exponent. Source: thesis Section 3.1, Eqs. 3.1 and 3.6.

Which flights contribute at each lag?

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All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

Which flights contribute at each lag?

Before fixing a population, look at what happens without one. At each lag we average over every flight that can contribute: a flight enters at lag τ if at least one of its retained segments satisfies $\tau \leq 0.8 T_s$. At least a fifth of that segment then remains available for origins, so m_f still averages over a sizeable stretch of the flight rather than a handful of displacements. Left: the equal-flight TA-MSD of all available flights. Trajectories sit on the common 10-second grid, so origins shift in 10-second steps; the plotted lags are a log-spaced subset. Shading shows pointwise 90 percent bootstrap intervals. Right: the number of contributing flights relative to the count at ten seconds.

Up to a few thousand seconds almost every flight contributes; from about 6,000 seconds the count falls steeply. Paragliders go from 155,085 flights at ten seconds to 46,273 at 10,000 seconds and 27 at 32,690 seconds; hang gliders from 6,060 to 2,326 and then 20 at 24,740 seconds. The dotted line marks a tenfold loss. The curves end where fewer than 20 site-day groups still contribute. A group is the set of flights from one take-off site on one calendar day; the bootstrap resamples whole groups, so with fewer than 20 the confidence bands would not be meaningful.

The circled end of the MSD is where this loss acts. There the curve steepens: around 6,600 seconds the local slope of the all-flight curve gives H near 0.90 for paragliders, whereas the fixed cohort of the next slide gives about 0.70 at the same lag. Same estimator, different flights: the steepening comes from which flights are averaged, not only from how they move. An H fitted to the all-flight curve therefore

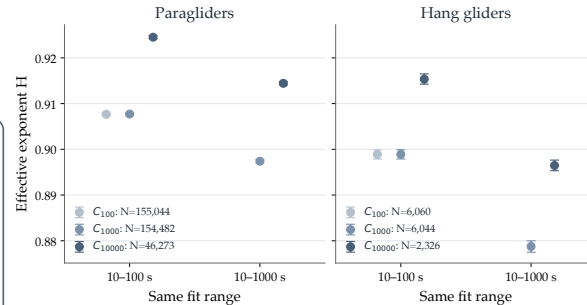
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.

About 2% in H buys the same flights at every lag over three decades.

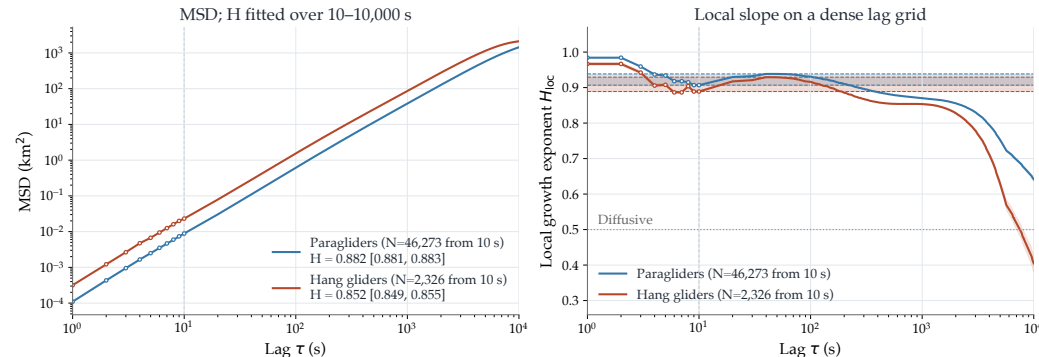
Can we afford a fixed population?

C_{10000} answers the three desiderata of the previous slide. We apply the admission rule once, at the largest lag: $10,000 \text{ s} \leq 0.8 T_s$ means segments of at least 12,500 seconds on the common grid. Even at the maximum lag, each segment keeps 2,500 seconds of origins, at least 251 on the 10-second grid. Only these segments enter, at every lag; shorter segments of an otherwise eligible flight are excluded throughout. Between 6,000 and 10,000 seconds, where the all-flight curve loses most of its flights, this cohort loses none. The cohort is exactly the set of flights still present at 10,000 seconds on the previous slide: 46,273 paragliders and 2,326 hang gliders, 30 and 38 percent of the flights available at ten seconds. So the sample stays of the same order as the full population, while covering three decades of lag. Pushing the maximum lag further would cost flights quickly: beyond about 16,000 seconds fewer than a tenth remain.

The price is a duration selection: long flights are not a random subset. The figure measures it by comparing nested cohorts over identical fit intervals, using paired bootstrap draws. The main cohort increases H by 0.0164–0.0177, about 1.8–2.0 percent. We accept this small, measured shift to keep the population constant across three decades of lag. Origins per flight still vary with lag. Source: thesis Section 3.1.3 and ch3_transport_report.json.

C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$H_{\text{PG}} = 0.882$ [0.881, 0.883] > $H_{\text{HG}} = 0.852$ [0.849, 0.855] over 10–10,000 s

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Displacement growth from 1 s to 10,000 s

The parent population is the original C10000 cohort, with the same selected long segments. Below ten seconds, a segment contributes when its native logging interval is no larger than the lag. Thus the contributing population grows from 33,450 paragliders and 690 hang gliders at one second to the full 46,273 and 2,326 at ten seconds. It stays fixed thereafter, through 10,000 seconds.

At short lags we use native origins and linearly interpolate endpoints when the lag is not a multiple of the logging interval, always within the selected segment span. This monotone admission rule differs from Figure 3.1: that figure uses the general population and requires exact native short-lag resolution, so its short-lag counts need not grow monotonically. From ten seconds, this slide uses the thesis's common ten-second grid and reproduces every original MSD point.

There are 277 evaluated lags, including every second from one to ten. Local H is half the log-log slope in a plus/minus 0.25-decade window, expanded to the nearest three lags when sparse. The one-second estimate uses one to three seconds. Slopes below ten seconds, and windows crossing that boundary, combine dynamics with changing sample composition. Lines connect computed estimates. In the right panel, each shaded horizontal band spans that discipline's local-slope range from its local minimum near ten seconds to its local maximum near 40–50 seconds.

The original 10–10,000 second fits and 1,000 paired site-day bootstrap draws are preserved: $H=0.882$ [0.881, 0.883] and 0.852 [0.849, 0.855]. Bands are pointwise 90 percent intervals. Sources: thesis Section

Which flight characteristics shape MSD and H ?

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Circuit type

Declared route

Open versus **closed**

Terrain / altitude

Proxy: launch altitude

Plains, **hills**, **low** and **high** mountains

Experience

Proxy: wing class

Beginners: EN A/B/C
Experts: EN D/CCC

Weather also matters, but we have no data with sufficient spatial and temporal resolution.

Marginal comparison

Example: open vs closed flights
across all altitude classes.

Stratified comparison

Example: open vs closed flights
within each altitude class.

What stratification controls

The altitude class is shared; weather and equipment can still differ between groups.

First compare groups overall, then within the same altitude class.

Which flight characteristics shape MSD and H ?

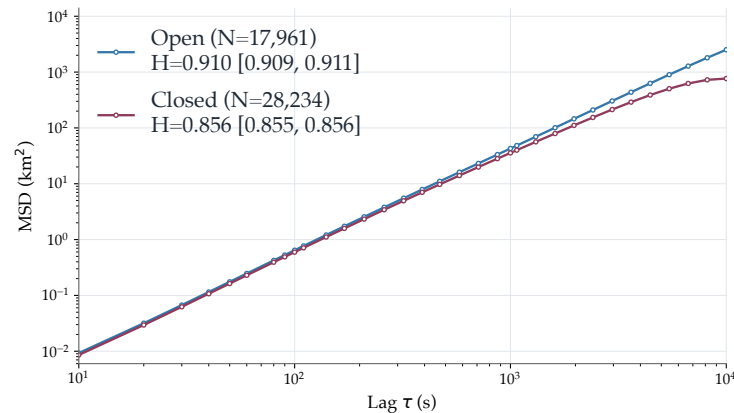
We examine circuit type, launch-altitude class and wing class. Launch altitude describes the starting environment, rather than the full terrain traversed. Wing class reflects equipment performance as well as a proxy for pilot experience.

A marginal circuit comparison pools altitude classes: open and closed flights can have different altitude compositions. Stratification compares open and closed flights within Plains, then within Hills, Low mountains and High mountains. The altitude class is fixed, while equipment, weather and the precise altitude within that class can still differ. The comparison is conditional on altitude class and still marginal over the remaining characteristics. Stratification does not require all flight conditions to be identical.

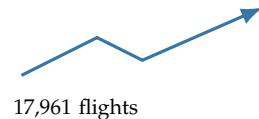
The next figures show pooled circuit and altitude contrasts, then examine circuit composition and its contribution to the pooled MSD. We then compare circuits within altitude classes: does the open-closed contrast persist, and does its size change? It persists but narrows in mountains, and the altitude ordering differs between open and closed circuits. Later comparisons examine regions and wing classes within selected altitude categories.

These are observational associations. Stratification accounts for the selected category, but alone does not isolate causal effects of circuit choice, terrain or equipment. Source: thesis Section 3.2.

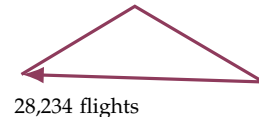
Open vs closed: how do MSD and H differ?



Open circuits



Closed circuits



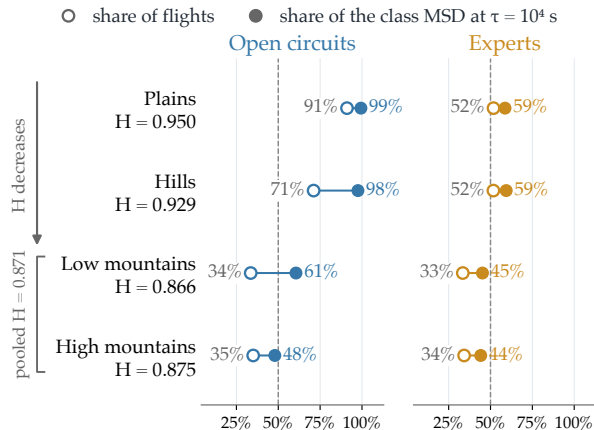
$H_{\text{open}} - H_{\text{closed}} = 0.054$ [0.053, 0.056] > 0 (90% paired interval)
Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

The catalogue describes the intended circuit, not a guarantee that the trimmed trajectory returns to its starting point. Closed routes include triangles and out-and-return flights. Pooling launch altitudes gives 0.910 [0.909, 0.911] for open routes and 0.856 [0.855, 0.856] for closed routes. Their paired difference is positive under the adopted bootstrap.

We fix the lag and average squared displacements over starting times along each segment. Suppose an ideal pilot reverses direction at minute 60, with the same speed on both legs. A two-minute interval from minute 59 to 61 spans the reversal and ends at its starting position; minute 20 to 22 covers straight flight. Thus even a short lag can capture a turn late in the flight. With equal leg times, $T/2$ and $T/3$ locate the first major turn along out-and-return and triangular routes. They do not define a minimum affected lag. Ten thousand seconds is the analysis limit; flight and leg durations vary. Testing the triangle hypothesis requires separating route types and examining actual leg durations.

The schematics illustrate geometry only. This marginal comparison pools launch environments and equipment, so its magnitude cannot rank the causal importance of route choice against terrain. Sources: thesis Section 3.2.1, ch3_conditional.json.



1 Different mix ○

Lowlands: more open circuits and more Experts, both with higher H .

2 Different weight ●

Group g , a circuit type or wing class, enters as flight share $\times$ MSD:

$$M_{2,a}(\tau) = \sum_g p_{g|a} M_{2,g,a}(\tau)$$

Low mountains: open flights are 34% of flights, 61% of the MSD at 10^4 s. Circuit type shifts the weights more than wing class.

Pooled H : Plains > Hills > mountains; the circuit and wing mix changes with altitude too.
An altitude-only comparison mixes terrain with circuit and wing composition.

Shares of all flights in each class; unclassified: circuit < 0.3%, wing up to 10%. Observed shares, no confidence bands.

Does the altitude ranking of H measure terrain alone?

Pooled over circuits and wings, H orders Plains 0.950 [0.948, 0.951], Hills 0.929 [0.926, 0.931], Low mountains 0.866 [0.865, 0.867] and High mountains 0.875 [0.874, 0.876]. The arrow on the left of the figure marks the fall from Plains to Hills to the two mountain classes pooled, 0.871 [0.870, 0.871]. The decrease is monotone only after pooling: High mountains sit slightly above Low mountains, and their nominal intervals do not overlap. Before reading this ranking as terrain, look at what each class contains.

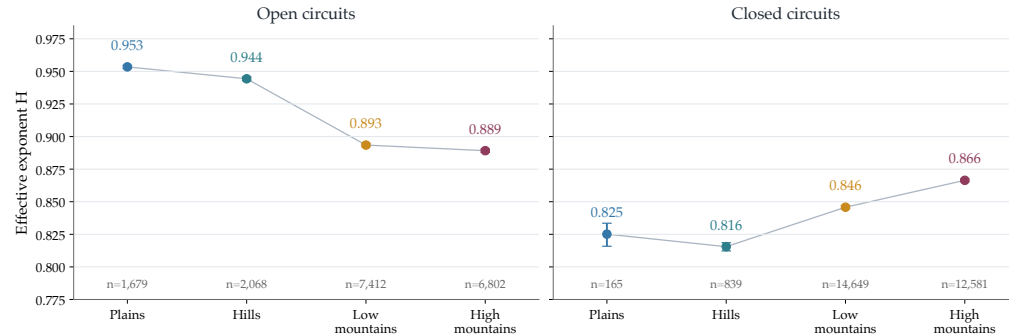
Point 1, hollow dots: Plains and Hills are 91 and 71 percent open circuits and 52 percent Experts; both mountain classes are about one third of each. Pooled, open circuits grow faster than closed ones, 0.910 [0.909, 0.911] against 0.856 [0.855, 0.856], and Experts faster than Beginners, 0.895 [0.893, 0.896] against 0.870 [0.869, 0.871] (slides 13 and 16).

Point 2, filled dots: in the left panel a group is a circuit type, open or closed; in the right panel it is a wing class, Expert or Beginner. Group g supplies $p_{g|a} M_{2,g,a} / M_{2,a}$ of the class MSD. At 10,000 seconds in Low mountains, open flights reach 1,814 km² against 596 km² for closed ones: 33.5 percent of the flights supply 60.6 percent of the MSD. For Experts the shift is smaller, 33.4 to 45.2 percent.

The MSD share w_g is not fixed along the lag. At 10 seconds it is close to the flight share; in Low mountains the open share of the MSD rises from 34.6 percent at 10 seconds to 60.6 percent at 10,000 seconds, a gain of 26 points. With local exponents k defined by $M_{2,g,a} \propto \tau^{2k}$, $d \ln w_g / d \ln \tau = 2(k - k_a)$.

Does the open–closed H gap change with altitude?

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Open: lowland H is higher. In mountains, each transition must head for known lift, so the flight moves out less directly.

Closed: mountain H is higher. In plains, pilots can turn back sooner and more often; in mountains, even the way home must follow the lift.

Plains: thermals turn up here and there. Mountains: routes must follow ridges and slopes.

Open > closed in every class; mountain orography constrains both, and the gap narrows.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Does the open–closed H gap change with altitude?

Read this figure in two directions. Comparing circuits within each altitude class gives open H greater than closed H in all four classes. The paired differences are 0.128 [0.120, 0.138] in Plains, 0.129 [0.125, 0.132] in Hills, 0.048 [0.046, 0.049] in Low mountains and 0.023 [0.021, 0.024] in High mountains. All four nominal 90 percent intervals are above zero. The circuit ordering survives altitude stratification, so different altitude composition alone cannot explain it. This is a consistent conditional association; its size still depends on altitude.

Comparing altitudes within each circuit reveals the reversal. Lowland-minus-mountain H contrasts are positive for open routes (0.051–0.064) and negative for closed routes (magnitudes 0.021–0.051). All eight paired intervals exclude zero. There is no general mountain penalty in H .

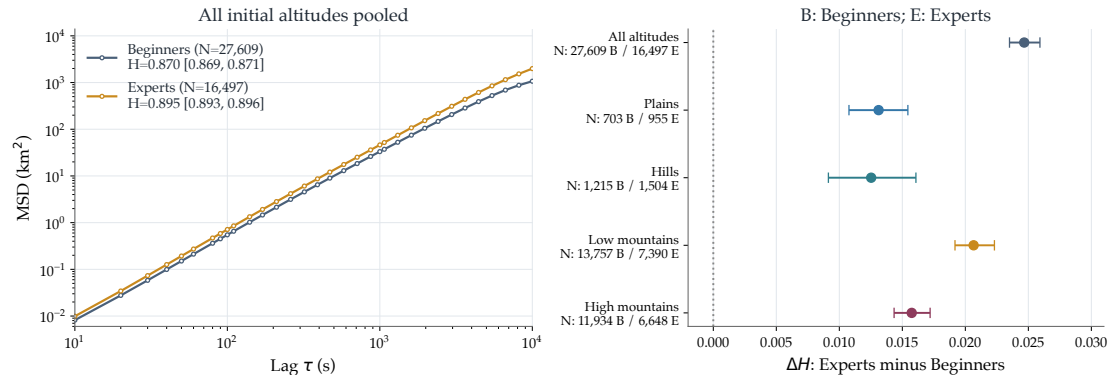
The two findings combine into a narrower circuit gap in mountains, as quantified by the four paired estimates above. Our interpretation is that orography constrains how pilots carry out either task. In mountains, usable thermals are localised on sun-facing slopes and along ridges, so each transition must be planned towards places where lift is known to form. In plains, lift use is spread more evenly in space and in height, with fewer clusters: a pilot meets thermals here and there along the way. Open circuits are therefore somewhat less open in mountains. Glides bend towards the known lift zones, so outward displacement, and with it the MSD, grows more slowly than in plains. Closed circuits show the opposite effect. In plains and hills the pilot is freer to turn back, sooner and more often, so

Does the expert–beginner H gap change with altitude?

16

EconophysicsLab offsite 2026 | Slide 16

Does the expert–beginner H gap change with altitude?



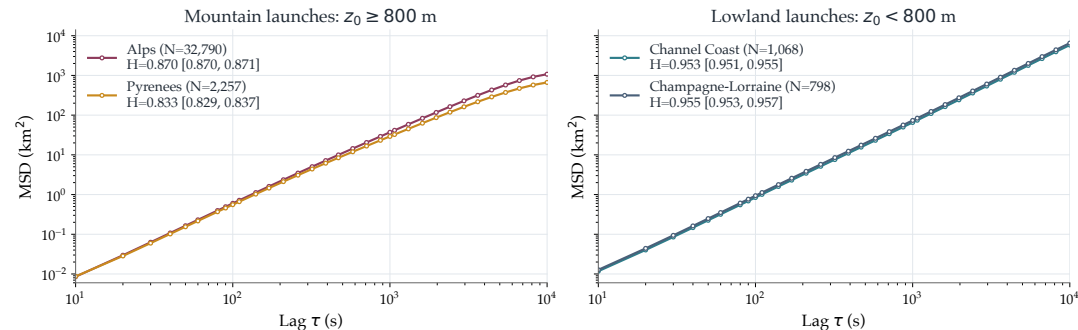
Higher H for experts in every class; the gap depends on altitude.

A constant speed advantage changes MSD amplitude, not H .

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

The Beginners group contains 27,609 flights and the Experts group 16,497; 2,167 flights are excluded from this comparison. These are the operational groups defined on slide 12. Individual experience is not observed: an experienced pilot can fly a lower-class wing, and the same pilot can contribute to different groups on different flights. The pooled exponent difference is 0.025 [0.024, 0.026]. Within altitude classes the differences are 0.013 [0.011, 0.015], 0.013 [0.009, 0.016], 0.021 [0.019, 0.022] and 0.016 [0.014, 0.017], all with paired intervals excluding zero. The Low mountains contrast exceeds the others in the paired interaction tests. A constant speed multiplier alone changes MSD amplitude, not its log–log slope, so the difference concerns the lag dependence too. The altitude-dependent gap is an equipment-by-altitude interaction already tested with paired contrasts. Each cell still pools circuits and weather. Circuit and equipment contrasts both depend on altitude, motivating the joint model with interaction terms discussed in the future-work section and the appendix. Source: thesis Section 3.2.4, ch3_conditional.json and ch3_conditional_equipment_table.tex.

Within altitude groups, how does H differ by region?



Mountain hypothesis

Slope-linked lift ties routes to ridge and valley geometry.

Lowland hypothesis

Roads and dark fields may supply broadly distributed thermal triggers.

ΔH : Alps–Pyrenees 0.037 [0.033, 0.041]; lowlands -0.002 [-0.005 , 0.001].
Similar lowland H ; Pyrenees may constrain routes more than the Alps.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

The regional selections hold broad launch altitude fixed: at least 800 m in the Alps and Pyrenees, below 800 m in Channel Coast and Champagne-Lorraine. Legends give the selected flight counts. Alps H is 0.870 [0.870, 0.871] against 0.833 [0.829, 0.837] in the Pyrenees, with paired difference 0.037 [0.033, 0.041]. The lowland difference is -0.002 [-0.005 , 0.001]: their growth exponents are similar, although MSD amplitudes differ.

The identity of the plain may matter less for H when lift-use locations have comparable spatial structure. Two appendix slides map where classified climbs occurred. Those maps show observed use, not the atmospheric distribution of thermal sources. Roads, asphalt and dark ploughed fields are candidate lowland triggers; comparable spacing and availability remain hypotheses.

In mountains, thermally heated slopes and wind-driven slope lift link usable lift to ridge and valley geometry. We interpret lower H and smaller long-lag MSD in the Pyrenees as consistent with stronger constraints on net progress than in the Alps. Alpine terrain may offer more connected route choices. H alone does not count accessible directions. The open shares in the two mountain regional cohorts are 32.1 and 30.9 percent, reducing the concern that a large circuit-mix difference alone explains the contrast. Equipment, weather and detailed launch altitude remain unmatched.

Sources: thesis Section 3.2.3; ch3_conditional.json; SSA, Lift Sources; FAA Glider Flying Handbook, Chapter 9.

How do climb locations change with height?



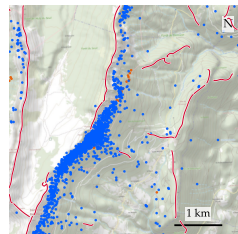
All dates and years pooled. Vilpelle climb crossings.

H2: Chartreuse, Alps

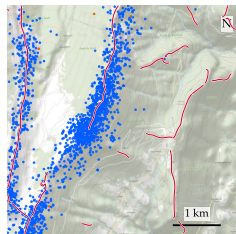
High mountains #2 / 15,840 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,856 crossings / 1,261 flights
PG: 1,249 HG: 12



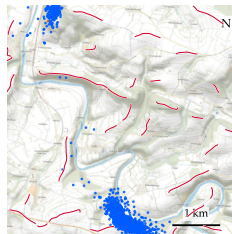
1,868 crossings / 1,452 flights
PG: 1,434 HG: 18

P1: Suisse Normande, Channel Coast

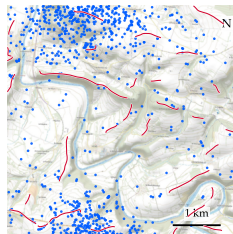
Plains #1 / 2,246 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,441 crossings / 734 flights
PG: 733 HG: 1



901 crossings / 574 flights
PG: 574 HG: 0

100 to 600 m: P1 climbs spread out; H2 climbs stay near slopes and ridges.

Observed lift use is less confined aloft in P1, supporting more spatial freedom for route choices.

The two five-kilometre viewer cells are H2, High mountains number 2 near Aulp du Seuil in the Chartreuse, Alps; and P1, Plains number 1 in inland Suisse Normande within the Channel Coast box. P1 contains local slopes despite its low starting-altitude class. Both disciplines and all available dates through June 2026 contribute, without seasonal or MSD-duration cuts.

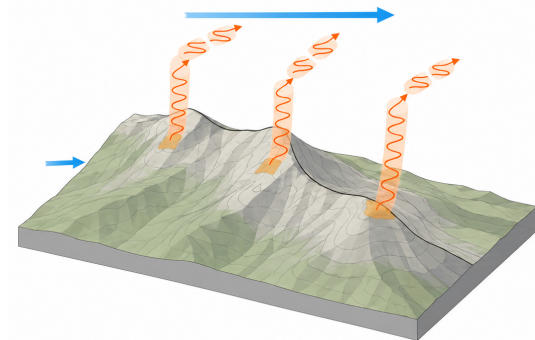
Heights are offsets from fixed references: 1756 m ASL in H2, based on five screened starts, and 185 m in P1, based on 1165. They are not local terrain clearance. Red crests are independently derived from IGN RGE ALTI at 25 m sampling with 50 m smoothing. These scale-dependent traces include secondary ridges and retain gaps at weak or ambiguous junctions. They do not move with the flight plane.

H2's two main bands at 100 m chiefly follow outer slopes of separate ridges: Lances de Malissard and Aulp du Seuil-Bellefont, enclosing the Vallon de Marcieu. They do not simply flank one central crest. In P1, crossings cluster near local slopes at 100 m and occupy a broader area at 600 m; H2 retains narrower terrain-associated corridors.

This is observed lift use, not tracked thermal centres. Possible slope lift and different flights and weather contribute at each height. Source distribution and wind advection are proposed explanations on the next slide; neither wind profiles nor their contribution to H are measured here. Sources: prepared viewer intersections: IGN Plan IGN and RGE ALTI. Licence Ouverte 2.0.

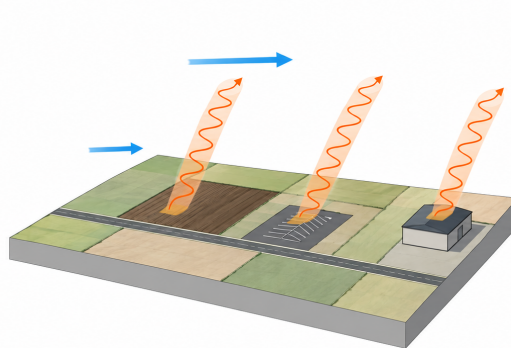
Mountains: sources on heated slopes

Possible shelter below crests; stronger shear aloft.



Lowlands: separated heated surfaces

Dry field, asphalt, warm roof; drift from the base.



Distributed sources and variable wind could broaden lowland crossings.

Mountain slope sources and possible shelter can keep lift use closer to terrain corridors.

Orange: rising air; blue: wind. Illustrative profiles, not measurements. Sources: FAA, Ch. 9–10; SSA, Lift Sources.

Why do lowland climb locations spread aloft?

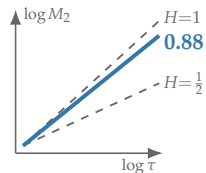
Distributed heat sources and wind advection could explain the stronger broadening between 100 and 600 m in P1. Slope-associated sources and possible shelter could preserve a narrower mountain footprint. Mountain sources can form on heated slopes and move upslope before detaching; they need not originate on crests. Dry fields, asphalt and roofs illustrate separate potential lowland sources, not guaranteed thermals. Orange is buoyant rising air, blue is horizontal wind. Helices indicate ascent, not measured rotation; fragments illustrate possible shear disruption.

Sheltered slopes with stronger flow above the crest are one possible mountain profile. Exposed lowland sources may drift from lower heights. Neither profile is measured or universal: valleys can channel strong wind, and inland P1 is not necessarily coastal.

For a steady idealised plume, $\Delta r_h \simeq (u_h/w)\Delta z$, where w is upward air speed, not glider climb rate. Shared wind and rise speed translate a source pattern; variability across dates can broaden pooled crossings. Matched climbs and wind profiles are needed to test this.

Terrain guidance and clearance could interrupt open routes' outward progress. Longer legs or later returns could weaken closure's signature in closed routes. Both may narrow the circuit gap; route timing remains untested. Sources: FAA Glider Flying Handbook, Chapters 9–10; SSA, Lift Sources; Schroeder and Buck, Fire Weather, Chapter 6; thesis Section 3.2.2.

Scaling



Strongly superdiffusive

$H = 0.88$ for paragliders, 0.85 for hang gliders, fitted over 10s – 10^4s on the same flights at every lag.

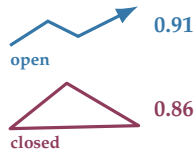
Terrain



Mountains narrow the gap

In mountains, open-route H falls and closed-route H rises. Altitude alone mixes terrain with route and wing mix.

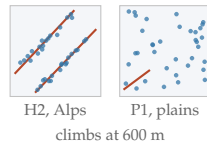
Route and wing



Open routes spread faster

Open exceeds closed in every altitude class (+0.054 pooled). Experts exceed beginners, by less (+0.025).

Lift geography



Mountain lift hugs ridges

From 100 to 600 m, H_2 climbs keep to slopes and ridges; P_1 climbs spread out. Observed lift use; causes untested.

Strongly superdiffusive in every group; route and terrain interact, wing adds a small shift

Shifting hypothesis: terrain constrains routes through where pilots can find lift.

Take-home messages

Four findings, one per cell; the band gives the synthesis. What we are still working on follows on the next slide.

Scaling. Over 10 to 10,000 s the fixed C_{10000} cohort gives $H = 0.882$ for paragliders and 0.852 for hang gliders, between diffusion at one half and ballistic motion at one. The same flights enter at every lag, so the exponent is not produced by long lags keeping only long flights. The pictogram is schematic: the reference lines are the diffusive and ballistic slopes, the blue line has the fitted paraglider slope. H summarises the fit range: the local slope stays near 0.85–0.94 up to about 2,000 s and then declines, to about 0.64 for paragliders and 0.41 for hang gliders at 10,000 s (slide 11, right panel).

Route and wing. Pooled over altitude classes, open circuits reach 0.910 [0.909, 0.911] and closed circuits 0.856 [0.855, 0.856], a paired difference of 0.054; open exceeds closed within every altitude class. Experts exceed Beginners in every class too, by 0.025 pooled and 0.013–0.021 within classes.

Terrain. The miniature plots the eight circuit-by-altitude exponents of slide 15. Open routes have higher H in the lowlands, closed routes in the mountains, so there is no general mountain penalty. The open–closed gap is 0.128 [0.120, 0.138] in Plains and 0.023 [0.021, 0.024] in High mountains. Because Plains are 91 percent open circuits and the mountain classes about one third, a ranking by altitude alone mixes terrain with route and wing composition (slide 14). Regionally, the two lowlands have similar H and the Pyrenees sit below the Alps by 0.037.

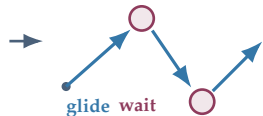
Segmentation



Revise Vilpellet's HMM

Built on ≥ 1 Hz tracks; misses some of our climbs. Next: continuous features in physical time.

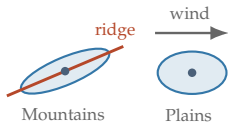
Stochastic model



Describe the observations

Candidate: glides alternate with waits (search + climb), built from measured glides, turns and waiting times.

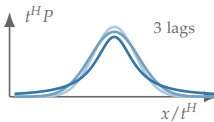
Anisotropy



Ridge axis or wind?

Mountains: chain axis. Plains: weaker, mean wind. Preliminary observations.

Distribution scaling



Do distributions collapse?

Collapse and moment tests: mixed so far. Power-law MSD alone does not establish self-similarity.

Work in progress: resolving phases and directions

The current Vilpellet segmentation is useful and already supplies the thermal-plane examples. The source study retained trajectories at a minimum frequency of one hertz; its appendix uses binary trajectory features. That domain does not validate transfer to our slower loggers. Visual inspection also suggests some climbing or lifting stretches are missed. These are diagnostics, not a completed accuracy benchmark. The proposed replacement remains an HMM with discrete glide, search and climb states, but continuous-valued observations such as vertical speed and turning measures. Features must be defined in physical time and state persistence must account for sampling interval. Merely replacing binary emissions with Gaussians does not solve cadence dependence. Validate on reviewed trajectories and deliberately downsampled high-frequency tracks.

The top-right cell states the goal: a stochastic model able to describe the empirical observations. The candidate has two phases: glides generate displacement, and each contiguous search and thermalling block becomes one waiting episode. Estimate glide lengths, durations, directions and turning-angle memory, and total waiting times without assuming independent search and climb durations. Position is held fixed during waiting, which neglects local search motion and thermal drift. Monte Carlo simulations with the observed sampling then test MSD, displacement distributions, anisotropy and temporal correlations. Details are in the appendix.

For anisotropy, estimate the displacement tensor at each lag and compare its main axis with local ridge

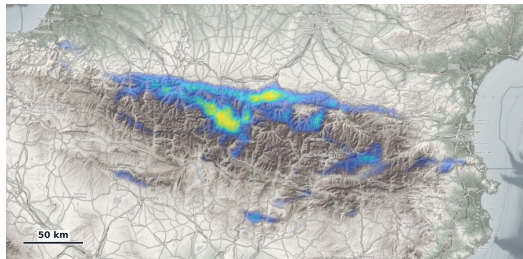
Thank you
for your attention

Thank you

Thank the audience and open the discussion. The regional climb-map construction, quantitative factorial example, solo-versus-group feasibility check, gliding-and-waiting model, velocity-memory definitions and bootstrap method follow as backup material.

Pyrenees: full chain

Atlantic to Mediterranean



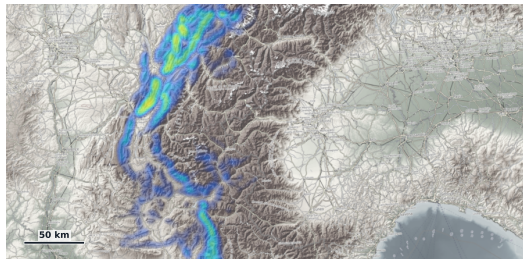
2,257 flights: open 698 (30.9%), closed 1,556 (68.9%)

$H = 0.833$

Share of flights with a climb

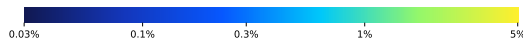
Western Alps

Matched 450 × 220 km area



32,790 flights: open 10,533 (32.1%), closed 22,212 (67.7%)

$H = 0.870$



Pyrenees: a narrow northern band. Alps: more branching climb areas.

Similar open shares make terrain guidance plausible, but the maps do not establish causation.

C10000 paragliders; all dates and heights pooled. Equal 450 × 220 km views and colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

Where do mountain climbs concentrate?

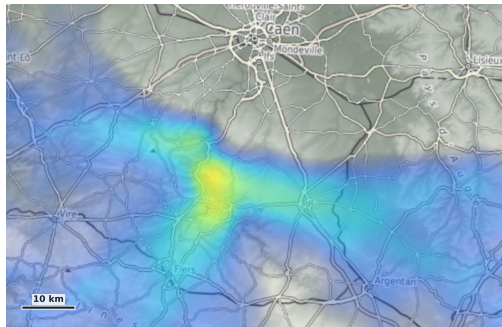
The Pyrenean map spans the chain from its Atlantic to Mediterranean ends, 450 by 220 km in Lambert-93. The western Alpine map covers the same physical area, but not the entire Alpine chain. The topographic base is lightly desaturated to keep its relief and ridge lines visible under the colour layer. The coloured field is observed climb use, not a census of all atmospheric thermal sources. In this archive, the Pyrenees have a strong elongated high-use band along the northern slopes, while the Alps show a main band with more branches through other valleys and ridge systems.

Both maps use the same C10000 paraglider cohorts as the regional H comparison: 2,257 Pyrenean flights (698 open, 1,556 closed, 3 unclassified) and 32,790 Alpine flights (10,533 open, 22,212 closed, 45 unclassified). The open shares are 30.9 and 32.1 percent, so the circuit mix is broadly comparable. H is 0.833 versus 0.870; the paired difference is 0.037 [0.033,0.041]. Similar circuit proportions do not remove weather, equipment, launch-location or route-selection differences.

No date, season or climb height is excluded. A flight counts once per 1 km cell with at least one own-HMM climb fix. Counts are divided by the regional flight number and smoothed by a 2 km Gaussian for display. Both panels use the same 0.03 to 5 percent logarithmic colour scale; low values fade to transparent. The OpenTopoMap background derives from OpenStreetMap and SRTM independently of flight coordinates. A more branched Alpine pattern is compatible with greater route freedom, but the map does not directly measure available lift or accessible directions and cannot explain H causally. See

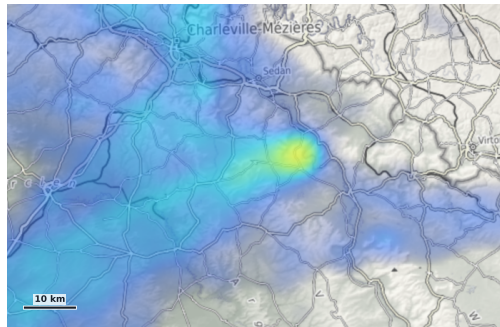
Channel Coast

1,068 flights, $H = 0.953$, 95.5% open

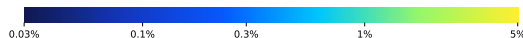


Champagne-Lorraine

798 flights, $H = 0.955$, 92.9% open



Share of flights with a climb



Both windows contain broad patches of climb use, with no single mountain-like crest corridor.

Their similar H values are compatible with weaker spatial constraints in lowlands.

C10000 paragliders; all dates and heights pooled. Matched 100×65 km lowland views; shared colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

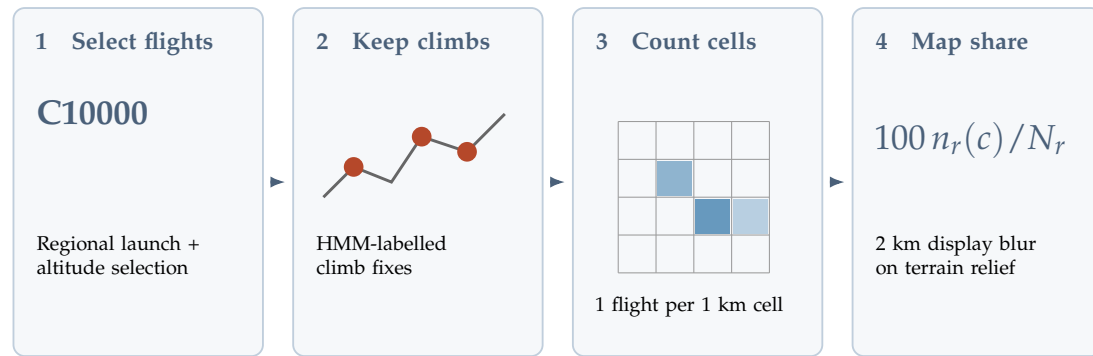
How broadly are lowland climbs distributed?

These matched 100 by 65 km views sample active portions of the Channel Coast and Champagne-Lorraine regional boxes. The first is inland Suisse Normande, not the immediate shoreline. The maps use the same C10000 paraglider population, own-HMM climb label, 1 km unique-flight grid, 2 km display smoothing and colour range as the mountain maps. There is no seasonal or height-plane restriction. Channel Coast has 1,068 flights, 1,020 open and 48 closed; Champagne-Lorraine has 798 flights, 741 open and 57 closed. Their open shares are 95.5 and 92.9 percent. Their fitted H values are 0.953 and 0.955, with a paired Channel-minus-Champagne difference of -0.002 [-0.005,0.001].

Both lowland examples occupy broad areas. They are not spatially uniform: visible hotspots remain, especially near active flight corridors. The maps support the weaker statement that climb use is less confined to a narrow mountain-crest network in these examples. More spatial freedom is a possible link to the larger open-minus-closed H gap in Plains and Hills shown earlier, but these plots do not measure available lift or prove that mechanism. Similarly, two matching lowland H values do not establish the same result for every plain or hill region. Unequal launch-site exposure, weather, wind and route choices remain alternatives.

The OpenTopoMap relief and contours are independent of flight coordinates. The method and limitations are listed in the appendix. Sources: ch3_conditional.json; render_regional_climb_density.py; OpenTopoMap (CC-BY-SA).

One chain for every map: the regional H cohort, with all dates and climb heights pooled.



Colour shows observed climb use, not every available thermal or a causal effect on H .

Source: FFVL C10000 tracks, own-HMM phase labels; OpenTopoMap relief (OSM contributors / SRTM, CC-BY-SA). Code and provenance: render_regional_climb_maps.py.

Appendix: construction of the 2D climb maps

The map population is the fixed 10–10,000 s C10000 paraglider cohort from chapter 3. A flight enters the Alps or Pyrenees sample if its cleaned launch lies in that geographic box and its initial altitude is at least 800 m. For Channel Coast or Champagne-Lorraine, the launch altitude is below 800 m. This reproduces the region memberships of the H comparison. No year, month, season or height plane is selected: all available dated fixes of those flights contribute.

The own-HMM phase_points.parquet supplies the climb label and each cleaned fix's fixed-frame E , N and GNSS altitude. The exact inverse WGS84 transform reconstructs longitude and latitude; Lambert-93 gives metric eastings and northings. A flight contributes one count to a one-kilometre square if it has at least one classified climb fix there, even if it circles for a long time or records many fixes. Divide by the total number of C10000 flights in that region. Display smoothing uses a Gaussian with 2 km standard deviation. Values below 0.03 percent are transparent and values at or above 5 percent saturate on the shared logarithmic colour scale. The Pyrenees map spans the whole chain in a 450 by 220 km view. The Alps uses the same physical extent and shows a selected western portion of that much larger chain. Lowland windows are each 100 by 65 km. The latter are illustrative subregions, not maps of the entire lowland regional boxes.

OpenTopoMap supplies the independent topographic background: OpenStreetMap map data, SRTM relief and a CC-BY-SA cartographic style. The map tiles are warped to Lambert-93 before the flight-

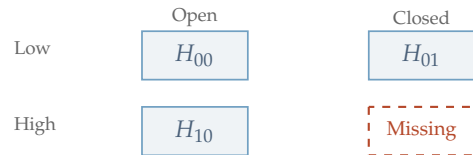
A: 4 altitudes C: 2 circuits E: beginners / experts $4 \times 2 \times 2 = 16$ cells

$$H_{ace} = \underbrace{\mu + A_a + C_c + E_e}_{\text{additive part}}$$

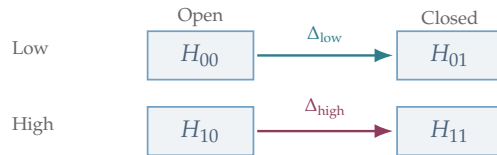
$$+ (AC)_{ac} + (AE)_{ae} + (CE)_{ce} + (ACE)_{ace}$$

Example: two altitude levels and two circuits; beginners / experts group held fixed.

OFAT: one combination missing



Full factorial: all four combinations



Interaction: $D_{AC} = \Delta_{\text{high}} - \Delta_{\text{low}} \neq 0$ means the circuit effect depends on altitude

Free coefficients:
6 additive
10 interactions

Sum-to-zero constraints

Appendix: factorial models and interactions

Fit H to each cell MSD over the same lag interval. Under sum-to-zero constraints, the additive model has six free coefficients: one mean, three altitude contrasts, one circuit contrast and one experience-group contrast. The ten interaction coefficients are three AC, three AE, one CE and three ACE. This saturates sixteen cells and supplies no residual error estimate by itself; weighting and uncertainty remain under review.

The diagrams fix experience group and code altitude and circuit as $x, z = 0, 1$. Rows are low/high altitude; columns are open/closed circuit. OFAT at (0,0) observes (0,0), (1,0), (0,1) and misses the high/closed cell. In $H_{xz} = \beta_0 + \beta_A x + \beta_C z + \beta_{AC} xz$, the interaction term is zero in those three settings, so its coefficient is undetermined.

The full grid supplies the fourth cell. The arrows give $\Delta_{\text{low}} = H_{01} - H_{00}$ and $\Delta_{\text{high}} = H_{11} - H_{10}$. Their difference is $D_{AC} = \beta_{AC}$. Additivity implies zero: the circuit effect is unchanged by altitude. A nonzero estimate requires uncertainty assessment. These cells are symbolic, not new measurements.

Archive comparisons are observational, not controlled OFAT. Binary group/solo doubles the table to 32 cells and adds its interactions. Reliable social labels, cell support and resampling for shared conditions are needed. No provisional factorial results are shown. Sources: NIST/SEMATECH, Sections 5.2.1.2 and 5.3.3.3; Mohr, Wilson and Freund (2022), Sections 9.3 and 9.7; thesis Section 3.2.5.

Appendix: is a solo versus group study feasible?

What a group label requires

Flights in the same small cell, close enough in time to see each other: visual contact, about 15 minutes apart at most.

Then compare transport within shared altitude, circuit and wing-class settings.

$$\Delta H_{a,c,e}^{\text{social}} = H_{a,c,e,\text{group}} - H_{a,c,e,\text{solo}}$$

What the archive gives

Best case: the most populated 5 km cell (Low mountains), busiest day, peak 15 min.

≤ 63 flights

22 July 2023, 13:15–13:30 Paris time.
An upper bound, before any split over 32 cells. Other cells are sparser.

Solo versus grouped flight was proposed as a factor in the joint factorial model. It requires reliable labels for simultaneous flight. A group here means flights that share a small spatial cell and are close enough in time for pilots to see each other, of the order of fifteen minutes. That rules out flights in the same cell on the same day but hours apart.

Even in the most favourable case the counts are far too small. Take the most populated 5 by 5 km cell, a Low-mountains cell, on its busiest summer day, 22 July 2023. In the peak quarter hour, 13:15 to 13:30 Paris time, at most 63 flights cross the cell, whether or not they climb there. This is an upper bound: a flight counts as present from its first to its last time in the cell, even if it leaves in between. Less populated cells hold fewer flights. Splitting by altitude class, circuit type and wing class, and then by group versus solo, divides such counts over 32 cells. The archive is therefore probably too sparse for this contrast, and the number is not a marginal shortfall.

Possible ways forward are a looser definition of co-presence, a coarser cell or a different data source with denser coverage, each at the cost of a weaker notion of visual contact. No group labels or social contrasts are reported.

With the current archive, a solo versus group comparison is most likely out of reach

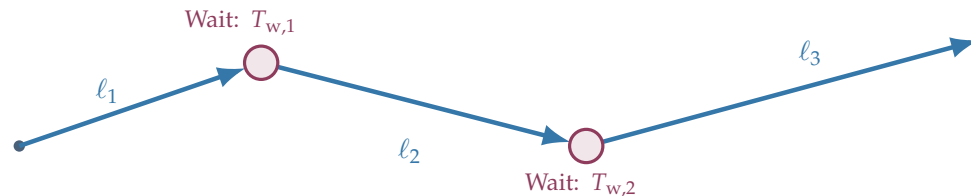
Appendix: gliding and waiting

Displacement: gliding

Duration T_g , length ℓ and direction θ .
Retain turning-angle memory between glides.

Waiting: search + thermalling

T_w : total time between successive glides.
Estimate one pooled waiting-time distribution.



Waiting neglects local search motion and thermal drift.

Monte Carlo: test MSD, displacement distributions, anisotropy and memory

Proposed model; schematic path.

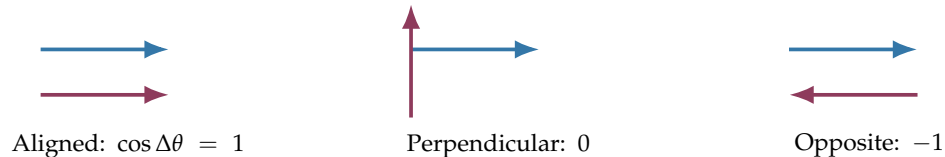
The proposed transport model has two phases: displacement during gliding, and waiting that combines search and thermalling. Vilpellet et al. report ballistic glides, subdiffusive search and slow horizontal drift during climbs. These observations motivate our reduction; they do not establish a fitted two-state model.

Retain three behavioural labels for segmentation, then merge each contiguous search/thermalling block into one waiting episode. Its duration includes all such intervals until the next glide, allowing repeated alternations or an absent search phase. Estimate total waiting times directly, without assuming independent search and climb durations.

Glides generate displacement over finite durations. Estimate their lengths, durations, directions and successive dependence, including turning-angle memory. In the simplest model, position stays fixed during waiting while time advances. This neglects local search motion and thermal advection: quantify the displacement error over the target scales. Reorientation between glides remains part of the model.

Simulate in two dimensions plus time, using the observed sampling and estimators. Test MSD, displacement distributions, anisotropy and temporal correlations on held-out flights. Matching one H is insufficient. Source: Vilpellet et al. (2026), Sections IV and VI, Figures 3–4.

$$\mathbf{v}(t) = s(t) \mathbf{e}(t), \quad \mathbf{e} = (\cos \theta, \sin \theta) \quad \Delta\theta = \theta(t + \tau) - \theta(t)$$



$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle s^2 \rangle} = \frac{\langle s(t)s(t + \tau) \cos \Delta\theta \rangle}{\langle s^2 \rangle}$$

$$C_\theta(\tau) = \langle \mathbf{e}(t) \cdot \mathbf{e}(t + \tau) \rangle = \langle \cos \Delta\theta \rangle \quad \text{Constant speed: } C_v = C_\theta.$$

Persistent mean drift can leave a nonzero baseline; also inspect centred velocity covariance.

C_v combines speed and direction; C_θ isolates directional agreement

Schematic vectors: blue at t , purple at $t + \tau$. Stationary normalisation; headings require nonzero speed. No empirical estimates.

Appendix: velocity and directional memory

Write horizontal velocity as speed times a unit heading. The dot product is the product of the two speeds multiplied by the cosine of their angular difference. Averaging and dividing by mean squared speed gives the displayed normalised velocity autocorrelation under stationary sampling. Aligned directions contribute positively, perpendicular directions give zero and reversals contribute negatively. Speed products weight those contributions, so velocity autocorrelation contains directional information but does not isolate it.

Correlating unit vectors removes speed weights: $C_\theta = \langle \cos \Delta\theta \rangle$. At constant speed, $C_v = C_\theta$. With varying speed they generally differ; speed and angular averages cannot be factorised without additional assumptions. Headings require nonzero speed.

Mean drift, including wind, can leave a positive long-lag baseline without memory in the fluctuations. The centred autocovariance is $K_v(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle - |\langle \mathbf{v} \rangle|^2$. It describes fluctuations around drift, whereas unit-vector correlation describes the original headings. State which convention is used.

For stationary velocities, writing $R_v(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle$ gives $M_2(\tau) = 2 \int_0^\tau (\tau - u) R_v(u) du$. This connects velocity dependence to displacement growth, but a fitted H does not specify that dependence uniquely. Use physical-time lags, within-segment pairs and consistent weighting. This slide derives identities; it does not report measured correlations.

Appendix: bootstrap confidence intervals

1 Group flights

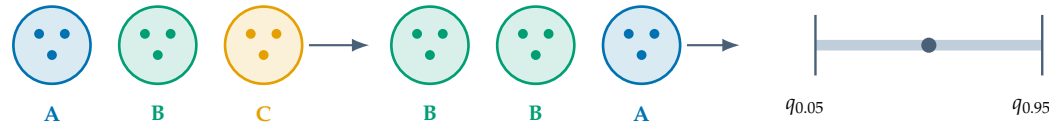
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

Schematic resample. Intervals are pointwise; they do not guarantee simultaneous curve coverage.

The reported confidence intervals use the current site–day cluster bootstrap. Each group combines flights from the same declared take-off site and calendar day. We sample groups with replacement, preserve all flights belonging to a sampled group, recompute the equal-flight estimator, and repeat this one thousand times. The same multiplicities are paired across lags and conditional comparisons. The 5th and 95th percentiles form nominal 90 percent intervals; for a curve they are pointwise, not a simultaneous confidence band.

Clustering aims to keep dependent flights together and make different groups approximately independent. The current partition is too fine: nearby sites and neighbouring days can share weather and other conditions, so flights in different groups can still be dependent. In the current cohort, 71.1 percent of paraglider groups and 81.3 percent of hang-glider groups are singletons. The bands are consequently likely to understate uncertainty; their nominal level should not be read as demonstrated coverage.

Validation with 50-km cells crossed with initial-altitude class and month of year gives standard errors for fitted H roughly four times the site–day value for paragliders and twice for hang gliders. Further aggregation changes the errors again; independence and uncertainty have not stabilised. These are diagnostic comparisons, not a validated replacement partition, and other chapter bands have not been recomputed with that scheme. All intervals in this presentation retain the original bootstrap convention. Sources: thesis Section 3.1.2; ch3_transport_grid_bootstrap.json and its generated values.

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.
parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

OpenTopoMap: regional relief and contours (OSM / SRTM, CC-BY-SA).
IGN / Géoplateforme: local Plan IGN backgrounds.
IGN RGE ALTI: local terrain and derived crests.

Locator and analysis

Natural Earth: France outline, public domain.
MSD, intervals and crossings: this thesis's prepared pipeline.
Code, figure scripts and recorded source provenance

Data and geographic sources

The FFVL archive supplies the observed trajectories and catalogue metadata. Paraglider and hang-glider collections have separate portals and season ranges, shown earlier. The pipeline applies the documented cleaning, selection and equal-flight estimators; generated figures are outputs of this thesis rather than independent external datasets. Thermal crossings in the five-kilometre cells come from the prepared viewer store. Regional climb-use maps come from the C10000 paraglider cohort and archived HMM phase fixes.

Regional maps use OpenTopoMap tiles, including OpenStreetMap map data and SRTM relief, warped to Lambert-93; the tiles and style are credited under CC-BY-SA. The background is desaturated and lightened for display. IGN Plan IGN is the coloured background of the separate small-cell maps. Approximate local crest lines are computed independently of flight positions from RGE ALTI terrain, requested at 25 m sampling for twelve selected five-kilometre cells with a 500 m processing buffer. The France locator uses Natural Earth country boundaries in the public domain.

Named summit points from IGN BD TOPO were tested earlier but are not the crest overlay currently displayed. The thermal-mechanism image is a schematic, not an atmospheric measurement. Source hashes, extraction parameters and figure scripts are recorded with the presentation and thesis.

Vilpellet, Darmon & Benzaquen (2026). *From Random Walks to Thermal Rides: Universal Anomalous Transport in Soaring Flights*. arXiv:2601.01293.
Transport, HMM phase segmentation and phase-duration benchmarks.

NIST/SEMATECH. *e-Handbook of Statistical Methods*,
§5.2.1.2: One Variable at a Time; §5.3.3.3: Full Factorial Designs.
Mohr, Wilson & Freund (2022). *Statistical Methods*, 4th ed., §9.3, §9.7.

FAA (2024). *Glider Flying Handbook*, Ch. 9–10.
Soaring Society of America. *Lift Sources*.
Schroeder & Buck (1970). *Fire Weather*, Ch. 6: General Winds.

Savitzky & Golay (1964). Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry* 36, 1627–1639. doi:10.1021/ac60214a047.

Selected bibliography

Vilpellet, Darmon and Benzaquen provide the central transport and segmentation comparison. Their reported phase durations are a benchmark to test after our own segmentation is validated. NIST and the statistical-methods textbook support the distinction between controlled one-factor-at-a-time experiments and factorial models with interactions. The FAA, SSA and general-wind sources support possible lift and wind mechanisms, not a measured wind profile for either selected cell. Savitzky and Golay give the local-polynomial basis of the smoothing and derivative estimates. The main talk keeps the statistical results, physical interpretations and proposed mechanisms distinct. The full references, publication details and links are maintained in thesis/references.bib.